

Task 5: Exploratory Data Analysis (EDA)

Titanic Dataset Analysis

Internship: Elevate Labs Data Analyst

Prepared by: Shobhish Balakrishnan

Objective, Dataset & Tools Used

Objective

The objective of this project was to perform Exploratory Data Analysis (EDA) on the Titanic dataset to identify patterns, trends, missing values, and relationships using statistical analysis and visualizations

.

Dataset

- Titanic Dataset
- 891 Rows
- 12 Columns

Tools Used

- Python
- Pandas
- Matplotlib
- Seaborn
- Jupyter Notebook

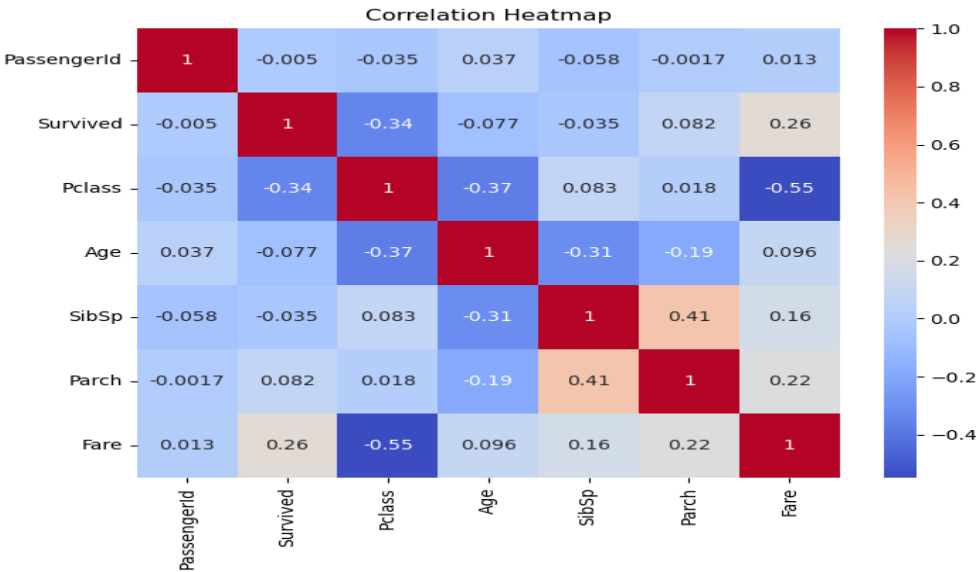
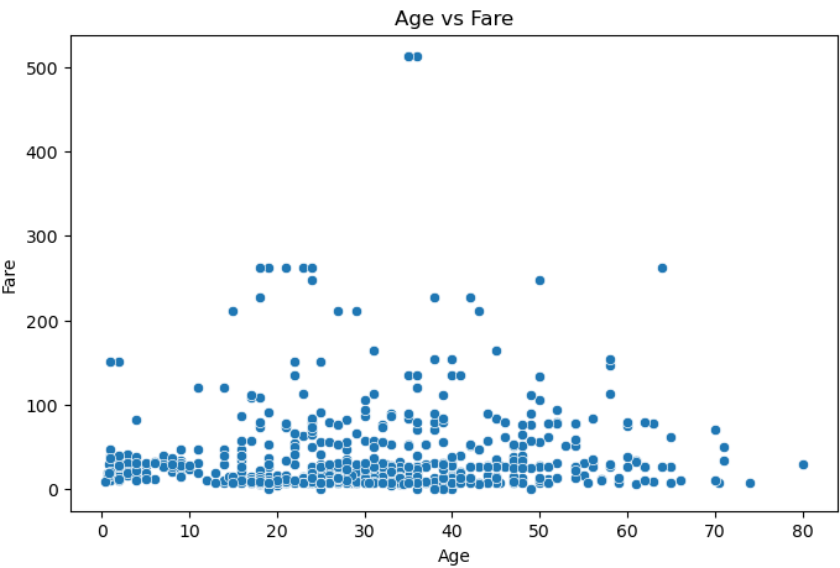
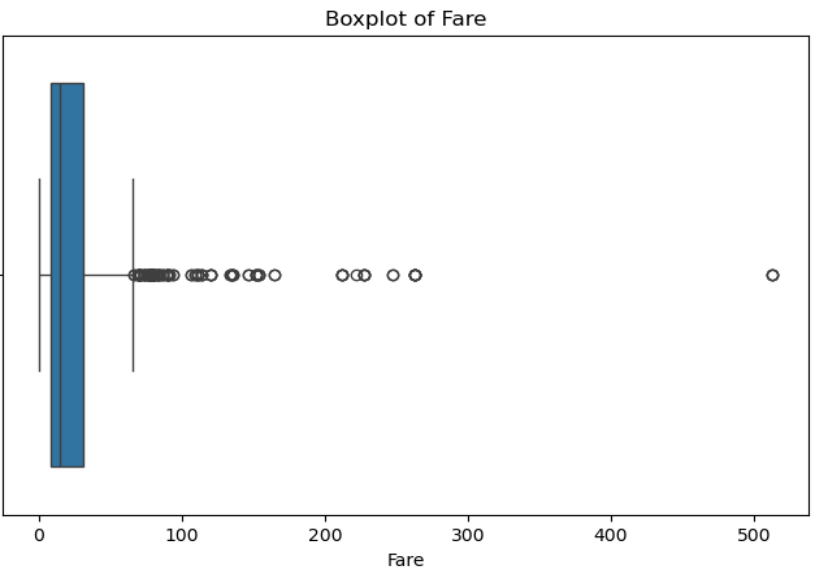
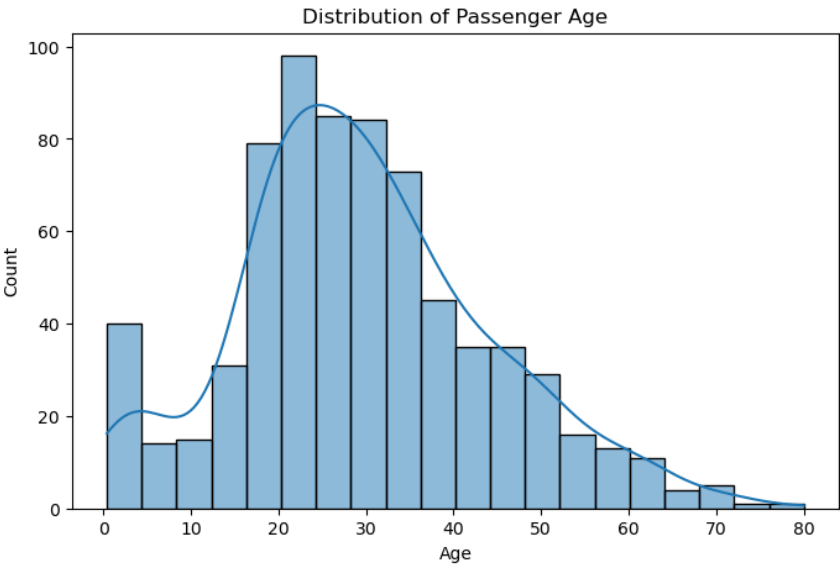
Analysis Performed

- Examined dataset structure using `.info()`
- Generated statistical summary using `.describe()`
- Identified missing values
- Used `.value_counts()` for categorical analysis
- Created Histogram to analyze age distribution
- Created Boxplot to detect fare outliers
- Created Scatter Plot to study Age vs Fare
- Generated Pair Plot for feature relationships
- Used Correlation Heatmap to analyze numerical relationships

Key Findings

- Dataset contains **891 passenger records**.
- Missing values found in **Age, Cabin, and Embarked**.
- Most passengers were **20–40 years** old.
- Majority of passengers traveled in **Third Class**.
- Male passengers outnumbered female passengers.
- More passengers **did not survive** than survived.
- Ticket fares contained several outliers.
- Correlation analysis showed mostly weak relationships.

Visualizations



Business Recommendations & Conclusion

Recommendations

- Handle missing values before predictive modeling.
- Treat outliers appropriately to improve analysis quality.
- Use visualizations to understand trends and patterns.
- Perform feature engineering for better machine learning performance.
- Apply EDA before any analytical or predictive project.

Conclusion

The exploratory data analysis successfully revealed important insights about passenger demographics, survival distribution, missing values, and variable relationships. The statistical summaries and visualizations provided a clear understanding of the dataset and established a strong foundation for future data preprocessing and predictive modeling.